proportional to the cube of the temperature. Find the value of $\gamma = (C_p/C_V)$

Ans: 3/2

7. Calculate the mean free path of air molecules at STP. The diameter of N_2 and O_2 is about $3 \times 10^{-10}\, m$

Ans: $\lambda \approx 9.3 \times 10^{-8}\, m$

8. A gas made of a mixture of 2 moles of oxygen and 4 moles of argon at temperature T. Calculate the energy of the gas in terms of RT. Neglect the vibrational modes.

Ans: 11RT

9. Estimate the total number of air molecules in a room of capacity of 25 m^3 at a temperature of 27°C with 1atm pressure.

Ans: 6.1×10^{26} molecules

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